1 THE END OF A MILLION YEARS OF FIRE

I was eight years old when I learned the world was ending.

It was 1974, in the Netherlands, and the oil crisis had hit. The Arab nations had cut off oil exports to punish the West for supporting Israel, and suddenly the grown-ups were scared. On Sundays, the highways stood empty. Private cars were banned. My parents and I walked in the middle of a highway that normally was full of traffic. That was kind of fun, but the silence was eerie, like everything held its breath.

At school, my teachers tried to explain what was happening. They showed us a poster that the government had distributed. I can still see it: the Earth depicted as a candle, burning from the top, its wax dripping away into space.

Verstandig met energie, it said. Be frugal with energy. Energy was finite. We were burning through it and once it was gone, it would be gone forever, and we would have burned up the earth.

The teachers also gave us a second reason to burn less. Global warming. The carbon dioxide was trapping heat. The ice caps would melt. The seas would rise.

But what unsettled me the most is that the teacher and my parents didn't have a solution. Civilization was doomed to end when we had burned through our fossil fuels. That was it.

There was no "but here's what we can do." No hope at the end of the lesson. Just fear, handed to an eight-year-old, to carry through his life.

I see that same fear now in my PhD students. I see it in the Extinction Rebellion activists who block traffic in Amsterdam. I see it in young people who have decided not to have children because, they say, the world is ending. The emotional toll of believing we are doomed is immense. It's a weight that bends people's lives.

It's also a lie and the reason I wrote this book.

But I'm getting ahead of myself. Before I can tell you why I no longer carry that weight, before I can explain why I believe we have found a way out, I need to show you where the fear came from. Not just for me, but for our entire species. Because we've been afraid of running out for a very long time.

SIDEBAR: What Is Energy?

Before we go further, let me offer a brief primer for readers who haven't thought much about energy since high school physics. Feel free to skip this if the terms are familiar.

Energy is the ability to do work. Anything that moves, heats, lifts, or transforms something else requires energy. When you climb stairs, your muscles burn energy. When a car accelerates, it uses the energy stored in fuel. When a light bulb glows, electrical energy becomes light and heat.

The basic unit of energy is the **joule**. One joule is roughly the energy required to lift a small apple one meter. It's a tiny amount, so we usually talk about thousands of joules (kilojoules) or millions (megajoules). The food you eat contains about 8,700 kilojoules per day for an average adult, or 10,000 to 14,000 kilojoules if you're moderately active, depending on body size and sex.

But here's where many people get confused: energy and **power** are not the same thing. Power is the *rate* at which you use energy, measured in **watts**. One watt means using one joule per second.

The unit on your electricity bill is usually the **kilowatt-hour** (kWh), which combines both ideas. If you run a 1,000-watt appliance (one kilowatt) for one hour, you've used one kilowatt-hour of energy. A typical European household uses around 3,000 to 4,000 kWh per year. (ODYSSEE-MURE, 2024) An American household uses closer to 10,000, largely because of air conditioning and larger homes. (EIA, 2024)

Some everyday examples: A toaster draws about 1,000 watts. (EcoFlow, 2024) Making toast for three minutes uses 0.05 kWh. An electric car charging overnight might draw 10,000 watts for five hours, using 50 kWh. A typical solar panel on your roof produces around 400 to 450 watts at peak output, generating perhaps 2 kWh on a sunny day. (EnergySage, 2025)

And what is **combustion**, the process that has powered human civilization for a million years? It's a chemical reaction that breaks the bonds in fuel molecules. Those bonds stored energy when they formed, originally captured from sunlight by ancient plants. When we burn wood, coal, oil, or gas, we're releasing that stored solar energy. The carbon in the fuel combines with oxygen from the air to produce carbon dioxide, water, and heat. This is why burning things always releases CO~2~: the carbon has to go somewhere.

All the energy debates of the coming chapters, whether about solar versus coal or batteries versus hydrogen, ultimately come down to different ways of capturing, storing, and releasing the same fundamental thing: energy. The units will reappear throughout this book. Now you have the vocabulary.

1.1 Fire: Humanity's First Technology

Control of fire is what made us human. This is not metaphor. Anthropologists estimate that our ancestors learned to use fire somewhere between one and two million years ago, and the consequences transformed our species. (Berna, 2012)

Fire gave us cooked food, which released more calories and allowed our brains to grow larger. (Wrangham, 2009) Fire gave us warmth, which let us colonize colder climates. Fire gave us light, extending the useful hours of the day and enabling social bonding around the hearth. Fire gave us protection from predators, a wall of flame between us and the darkness. Fire gave us the ability to shape materials: to harden spear tips, to fire clay into pottery, eventually to smelt metals into tools.

No other species controls fire. Chimpanzees, our closest relatives, can understand fire's behavior and calmly navigate around wildfires, but they have never been observed using or maintaining fire in the wild. (Pruetz, 2010) Fire is uniquely ours. It is the first technology, the foundation of everything that followed.

The ancients understood this. In Greek mythology, fire was so precious that it belonged only to the gods. Prometheus stole it and gave it to humanity, and for this crime he was punished eternally, chained to a rock while an eagle tore at his liver each day. The story appears across cultures in different forms: fire as gift from the divine, fire as transgression, fire as the beginning of human suffering alongside human progress.

Apart from fire being a sinful transgression, these myths also encode a second truth. Fire requires fuel which is scarce. And for 99 percent of human history, this meant that energy was a constant struggle. Every family needed wood or dung or peat to survive the winter. Every growing village meant forests pushed further back. (Nef, 1932) Every war could be partly measured in whose army ran out of fodder first.

Nature, for most of our existence, has been a fight for scarce resources. First food, then fuel. Everything that grows can be burned, and someone else might take it before you do. This is not neurosis. This is evolutionary inheritance. The scarcity mindset is hardwired into human psychology because scarcity was real for thousands of human generations and for over a billion years before that.

Consider how deep this goes. Children fight over toys before they can speak. Nations accumulate more weapons than they could ever use. We hoard, we compete, we assume that if someone else gains, we must lose. Economists call this zero-sum thinking, and they treat it as a cognitive error to be corrected. But for most of human history, it was simply an accurate description of the world. There was only so much wood in the forest. If your neighbor took it, you went cold.

1.2 Civilizations That Burned Out

The scarcity story isn't abstract. It left marks in the historical record. Civilizations rose on abundant energy and declined when that energy ran out.

Rome's expansion tracked its access to forests. The Italian peninsula was once densely wooded; by the height of the Empire, it was largely stripped. (DiLorenzo, 2023) Roman ships and Roman buildings consumed timber at rates the land could not sustain. The Empire pushed its logging frontier outward: to Gaul, to North Africa, to the limits of the Mediterranean world. When the trees grew scarce, so did the ships. When ships grew scarce, so did trade. The relationship is not simple, and I am not claiming that deforestation *caused* Rome's fall, but it definitely was an important part of the fall of Roman civilization.

A clearer case: the island of Hispaniola in the Caribbean, which today holds two nations, Haiti and the Dominican Republic. They share the same island, the same climate, the same starting conditions. Yet from the air, the border is visible: the Dominican side is green with forest cover; the Haitian side is brown and bare. (NASA, 2002)

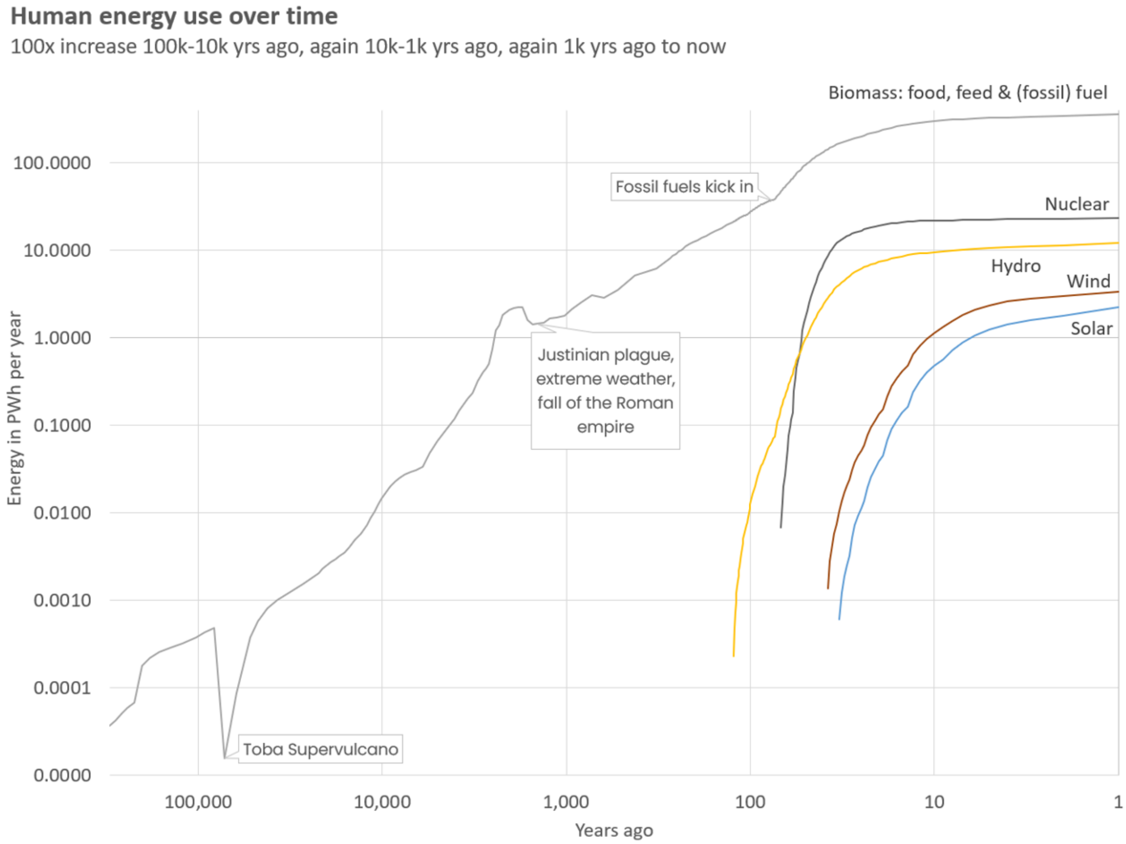
Haiti's deforestation is not ancient history. It happened largely in the twentieth century, driven by population pressure and the need for charcoal fuel. (WorldBank, 2019) Without trees, the soil washed away. Without soil, crops failed. Without crops, poverty deepened. I don't mean this as a morality tale about Haitian decisions; I see it as a case study in what happens when a growing population depends on biomass energy and has no alternative. Today, Haiti is one of the poorest nations in the Western hemisphere. (CFR, 2025) The Dominican Republic, which made different policy choices and retained its forests, is comparatively prosperous. Same island. Different outcomes. Energy scarcity, compounded over decades, reshaped a nation.

Or consider Medieval England. By the medieval period, England's woodland had already declined to around 15% of land area, with further losses in the sixteenth and seventeenth centuries for shipbuilding and charcoal production. (Sherren, 2020) Domestic heating, shipbuilding for the navy, and iron smelting for industry had consumed the trees. England faced a choice: decline or adapt.

Adaptation came in black lumps dug from the ground. Coal was known but despised, a dirty fuel that smoked and stank. No one *wanted* to burn coal. But with the forests gone, there was no alternative. (Nef, 1932) The transition was driven by desperation, not choice.

(Nef, 1932) And from that desperation came the foundation for the Industrial Revolution: a society that had learned to mine and burn fossil fuels because it had nothing else left.

SIDEBAR: Three Hundred Thousand Years in One Graph



This graph tells the energy history of humanity in a single image. Both axes are logarithmic, compressing vast scales into something visible.

For most of human history, our energy came from burning biomass: wood, dung, and eventually the crops we grew. Energy use has increased exponentially throughout human history. A logarithmic plot shows roughly hundredfold increases across major eras: from the pre-agricultural period to early agriculture, from early agriculture to pre-industrial civilization, and from the pre-industrial period to today.

Then fossil fuels kicked in. We discovered grandmother's savings under the mattress and started spending.

Notice the dip around 1,500 years ago: the Justinian plague and extreme weather that accompanied Rome's fall. (Gao, 2006) Energy use and civilization move together.

Notice what happens to nuclear: it rises, then flattens completely. It was a promise that didn't deliver and disillusioned my younger self.

Notice what happens at the far right: new lines appear. Hydro. Wind. Solar. They're still small compared to the grey biomass monster. Diminutive even. But they're growing. If you look at the logarithmic time axis you can see they are actually growing fast. And unlike everything else on this graph, they don't require burning anything.

Could solar and wind really replace the biomass Leviathan? That monster that grew a *millionfold* in the blink of an eye? This book shows that, yes, it can be done. Also how to do it, and why it might just be the best thing that ever happened to humanity.

1.3 Grandmother's Savings Under the Mattress

What we've been burning for the past two centuries is ancient biomass...

When we burn fossil fuels, we are burning hundreds of millions of years of accumulated sunlight. (Dukes, 2003) We are spending our grandmother's savings, the energy she stored under the mattress across geological ages. And we have been spending it very, very fast.

The spending spree began roughly 250 years ago, with the Industrial Revolution. (Britannica, 1998) In two and a half centuries, we have consumed a significant fraction of what took hundreds of millions of years to accumulate. Coal, which seemed inexhaustible to the Victorians, now has perhaps 100 to 150 years left at current consumption rates. (Worldometer, 2016) At current consumption rates, oil reserves might last until the 2070s, though the transition to alternatives may accelerate this timeline. Natural gas, perhaps a bit longer.

These estimates are imprecise, of course. We may find new deposits. Technology may extract more from existing fields. Demand may rise or fall. But the fundamental mathematics is not in dispute: fossil fuels are finite, and we are currently drawing them down roughly a million times faster than any natural process could replace them.

The fear of running out is older than I am. The nineteenth-century whale oil industry experienced its own depletion crisis as whale populations collapsed from hunting, causing prices to rise dramatically before kerosene emerged as an alternative. (York, 2017) The transition to kerosene, derived from petroleum, solved that crisis, only to create the oil dependence that shaped the twentieth century. The 1970s brought "peak oil" theories that predicted imminent depletion. Those predictions proved premature, but they weren't wrong in principle. They were only wrong about timing.

And timing, as anyone who has lived through an oil crisis knows, matters enormously. The OPEC embargo of 1973 didn't happen because oil ran out. It happened because a cartel of producing nations chose to restrict supply. The result was global economic shock, inflation, recession, and the car-free Sundays of my childhood.

When energy is scarce, those who control it wield enormous power. This is why the Gulf Wars were fought where they were. This is why petrostate dictators can fund wars and suppress their populations while the world looks away. We need what they have. That need shapes our foreign policy, our alliances, and our willingness to tolerate the intolerable.

I want to be clear about something here though: fossil fuels were miraculous. They lifted billions of people from poverty. They enabled the modern world, the food production that feeds eight billion humans, the medicine that extended our lives, the communication networks that connect us across continents. Anyone who lived through the twentieth century owes a debt to coal and oil and gas. We should be grateful.

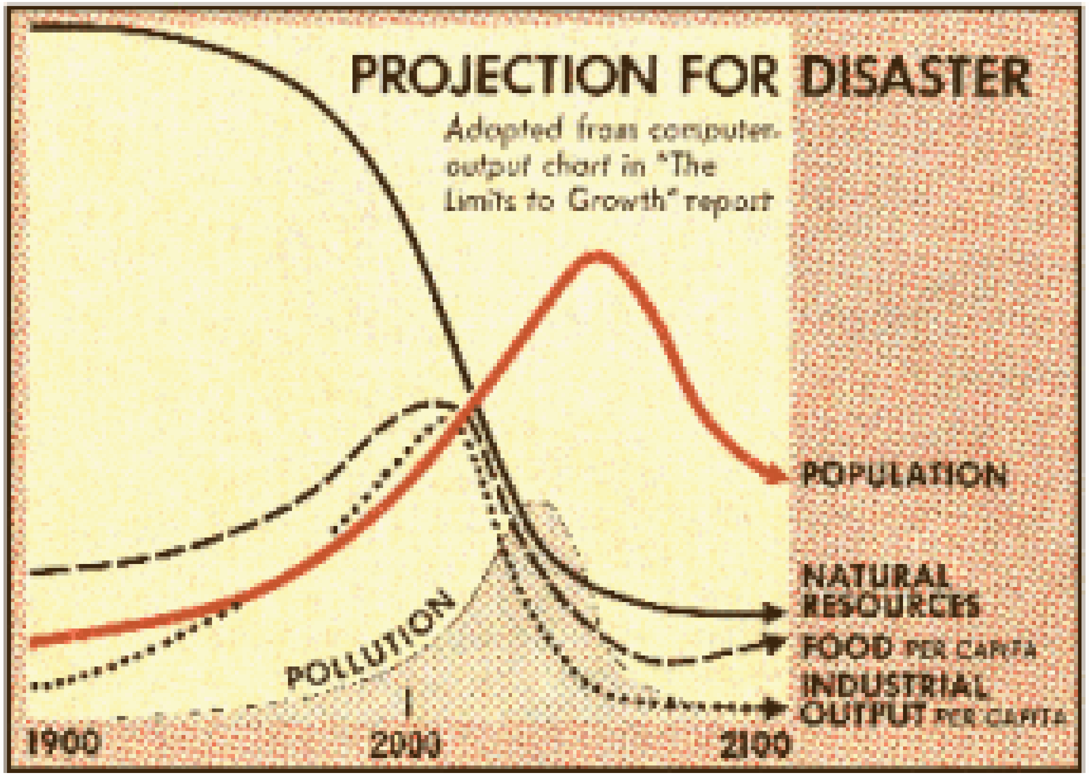
But gratitude does not change the mathematics. We cannot burn what doesn't exist. And we haven't even counted the true cost of the damage they cause, the carbon loading the atmosphere, the methane leaking from wells and pipelines, the health toll of air pollution in every city on Earth. The price we pay at the pump or on our utility bills is not the full price. There is a bill we have been deferring to the poor and to our children, and it comes due whether we acknowledge it or not.

1.4 The Limits to Growth

In 1972, just two years before I sat in that Dutch classroom watching the Earth-candle melt, a team of researchers at MIT had used a new approach called system dynamics and a brand new technology called a *computer* to see where our civilization was going. (Meadows, 1972) It's hard to imagine now, but a computer model was a bit of an awe inspiring novelty in those days. The computer model used stocks of resources: like the human population, soil for crops, fertilizer, oil reserves, pollution of the soil, et cetera. They traced the flows between them. They build their model, pressed "run" and watched what happened when exponential growth met finite limits. They documented the result in a book called *The Limits to Growth*. It sold four million copies. It terrified a generation.

They concluded humans were unable to see things like exponential growth and feedback loops before it was too late. Which was why human civilization kept using up resources at an alarming rate. An explanation next to the central graph of the report states: "Food, industrial output, and population grow exponentially until the rapidly diminishing resource base forces a slowdown in industrial growth. Because of natural delays in the system, both population and pollution continue to increase for some time after the peak of industrialization. Population growth is finally halted by a rise in the death rate due to decreased food and medical services." (Meadows, 1972)

The popular press expressed it even more vividly, like this quote from a 1972 Time magazine story (see also the accompanying graph): "The furnaces of Pittsburgh are cold; the assembly lines of Detroit are still. In Los Angeles, a few gaunt survivors of a plague desperately till freeway center strips . . . Fantastic? No, only grim inevitability if society continues its present dedication to growth and 'progress.'" (TIME, 1972)



The Limits to Growth model vividly demonstrated the power of exponential growth and feedback loops. An example of a feedback loop: more humans means more food means more humans. This feedback loop lead to an exponential growth of the human population so fast that it never fails to take our primate brains by surprise. And then we run out of food and the mass dying begins. Such predictions predate the use of computers of course. Malthus famously predicted famine would quickly hit this way in 1798. Tertullian went on about it almost two millennia ago.

And while Malthus was wrong, famine and collapse due to resource scarcity is hardly a historical novelty. It's what happened to Rome's forests. Trees are a stock. They grow slowly: that's the inflow. Romans cut them for ships, buildings, fuel: that's the outflow. For centuries, outflow exceeded inflow. The stock depleted. By the height of the Empire, Italy was largely stripped bare and most Romans probably knew, deep down, that the end was nigh. The same pattern explains Haiti, the cedars of Lebanon, the whales of the nineteenth century. Any time a civilization consumes a stock faster than it regenerates, the ending writes itself.

Still, most of their doom prophecies didn't come true yet. Why was that?

The first thing they underestimated, were feedback loops that pointed in the other direction. For example: once a stock gets scarce it becomes harder to find and less profitable. That slows down the exponential before it hits the wall. Scarcity also makes resources more expensive which makes it profitable to search for more resources and for substitutes. As economists like to say: the best cure for high prices, is high prices. Which is why we've managed to extend the deadlines from their business as usual scenario time and time again without hitting rock bottom yet.

The more fundamental thing they underestimated was that feedback loops could not only cause an exponential growth in problems, but also in solutions. I honestly think that is how humanity can transcend its current and immature use of finite resources. In using computer models to come up with exponentially growing solutions. That's what my efforts have been about the past two decades and in essence that is what this book is about: finding the virtuous feedback loops that will get us out of this mess with the speed of exponential growth.

1.5 The Emotional Weight

This is also a story about grief. There is a term for it now: eco-anxiety. Climate grief. Solastalgia, the distress caused by environmental change in one's home environment. (Albrecht, 2007) Psychologists study it. Therapists specialize in it. A generation of young people has grown up with the assumption that the future will be worse than the present, that the arc of history bends not toward justice but toward catastrophe.

I know this weight. It never dominated my life, but I carried it from that classroom in 1974: a kind of sombre soundtrack to any long term vision. In retrospect I think it was what plagued my teacher and I see it now in my students, bright young researchers who chose energy systems precisely because they want to help, and who struggle with despair when the numbers seem too large and the politics too broken.

I hear it when young people tell me they won't have children. Not because they don't want children, but because they believe it would be cruel to bring a child into a dying world. This is not a fringe position. Surveys show substantial minorities of young adults in wealthy countries expressing this view. (Hickman, 2021) The fear has reached deep enough to reshape the most fundamental human decisions.

Part of this grief comes from the assumption that nature is zero-sum. If humans prosper, something else must suffer. If we grow, forests must shrink. If we consume, the world must diminish. This assumption runs so deep that it feels like physical law rather than a model that might be questioned.

And there is exhaustion too: the exhaustion of caring, year after year, while emissions keep rising and conferences keep failing and politicians keep promising and breaking promises. At some point, the temptation of denial becomes overwhelming. If the problem is unsolvable, why keep suffering over it? Better to look away, to focus on what's in front of you, to leave the worrying to someone else.

I want to honor this. I understand where the despair comes from. The grief is earned. Anyone who tells you to simply cheer up has not understood the weight you're carrying. I can understand how genuinely believing that civilization is collapsing causes grief, denial, anxiety, and paralysis. The emotions follow from the beliefs.

But what if the beliefs are incomplete?

SIDEBAR: The Hunger Winter

In the winter of 1944-1945, the Nazis blockaded the western Netherlands in retaliation for a Dutch railway strike that had supported the Allied advance. (Lumey, 2007) Food shipments stopped. Fuel shipments stopped. As temperatures dropped below freezing, more than four million Dutch civilians faced starvation. (Lumey, 2007)

They called it the Hongerwinter, the Hunger Winter. People ate tulip bulbs and sugar beets. (Lumey, 2007) They burned furniture for heat. They walked for miles to trade family heirlooms for a few potatoes. Approximately 20,000 people died, most of them in the cities of the west, before Allied forces liberated the country in May 1945. (Lumey, 2007)

My parents were teenagers during the Hunger Winter. Neither talked about it often, but the experience shaped them. It shaped me.

What I inherited was not trauma but gratitude. Gratitude for warmth, for food, for the ordinary comforts that my parents' generation knew could vanish overnight. I also inherited a sense that security is not owed but earned, and that those who have it bear responsibility for those who don't.

I remember, around 1976, my mother packing our car for a family vacation to Hungary, then behind the Iron Curtain. But the car was not full of our vacation stuff. It was stuffed to the brink with clothing and supplies that our church had collected for Hungarian families. Through the church my parents had heard of people having less and for them it was just something you did. Mutual aid like that had kept people alive thirty years earlier.

This background shapes the voice you'll hear throughout this book. Dutch pragmatism: we built a country below sea level through collective action and long-term thinking and we are proud of that. Comfort as something that brings with it responsibility: if you are warm, help others get warm. Prosperity is not a sign of virtue, and poverty is not a sign of vice.

When I argue that abundant clean energy can lift billions from poverty, I feel that's both a pragmatic need--we will never get everybody’s cooperation otherwise--and a moral imperative.

1.6 What If We Were Wrong?

What if the scarcity story isn't the whole story?

I've spent the last two decades building models of energy systems. I've run the numbers on every scenario you can imagine. And I'm here to tell you something that I didn't believe myself until I saw it in the data: we are not running out of energy. We never were. We were only running out of the particular forms of stored ancient sunlight that was never the final answer anyway. The sun itself continues to flood the Earth with more energy every hour than human civilization uses in a year. (Lewis, 2006)

The question was never whether the energy existed. The question was whether we could capture it smartly. And in the past two decades, without most people noticing, the answer changed.

Two technologies now exist that break the pattern of the past million years. Solar panels convert sunlight directly into electricity. Wind turbines harvest the kinetic energy of moving air, itself driven by the sun's uneven heating of the atmosphere. Neither requires burning anything. Neither draws down a finite stock. Both tap into flows of energy that renew themselves every day, every hour, every moment the sun shines or the wind blows.

This is not wishful thinking. This is not idealism. The numbers are in, and they are astonishing. I will show them to you in the chapters ahead: how cheap solar and wind have become, how fast they are still falling in price, how the experts kept predicting they would fail and kept being wrong.

We thought we were running out. What if we were wrong?